

VODKA SALES ANALYSIS

**ESTIMATED PRICE ELASTICITY AND AD EFFECTIVENESS
USING REGRESSION MODELING.**

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1. Studying the Data

Study the data along with the data dictionary and write a summary about it.

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
1	BrandName	Char	9	\$9.	\$9.
2	Brand_ID	Num	8	BEST12.	BEST32.
11	Broad	Num	8	BEST12.	BEST32.
6	DollarSales	Num	8	BEST12.	BEST32.
5	Domestic	Num	8	BEST12.	BEST32.
15	Firstintro	Num	8	BEST12.	BEST32.
8	Mag	Num	8	BEST12.	BEST32.
16	Marketshare	Num	8	BEST12.	BEST32.
9	News	Num	8	BEST12.	BEST32.
10	Outdoor	Num	8	BEST12.	BEST32.
7	PricePerUnit	Num	8	BEST12.	BEST32.
12	Print	Num	8	BEST12.	BEST32.
13	Tier	Num	8	BEST12.	BEST32.
14	TotalMinusSales	Num	8	BEST12.	BEST32.
4	TotalSales	Num	8	BEST12.	BEST32.
3	Year	Num	8	BEST12.	BEST32.
17	YearID	Num	8	BEST12.	BEST32.

The SAS System

Obs	BrandName	Brand_ID	Year	TotalSales	Domestic	DollarSales	PricePerUnit	Mag	News	Outdoor	Broad	Print	Tier	TotalMinusSales	Firstintro	Marketshare	YearID
1	Absolut	15	1995	3000	0	352040	117.3467	25713.7	291.4	332.7	1	26005.1	1	60840	0	0.1339226	3
2	Absolut	15	1996	3340	0	370072	110.8	25658	177.6	625	1	25835.6	1	60532	0	0.1420068	4
3	Absolut	15	1997	3440	0	381152	110.8	27013.9	201.7	515.7	1	27215.6	1	60464	0	0.1467577	5
4	Absolut	15	1998	3630	0	419737	115.63	27617	308.2	1420.4	161.9	27925.2	1	60306	0	0.1461059	6
5	Absolut	15	1999	4050	0	468302	115.6301	30605.3	258.1	1416.8	70.3	30863.4	1	59918	0	0.1612839	7
6	Absolut	15	2000	4605	0	560025	121.6124	33971.3	533.4	1336.7	143.1	34504.7	1	59395	0	0.173368	8
7	Absolut	15	2001	4379	0	556550	127.0952	28559.8	231.3	1999	134.2	28791.1	1	59653	0	0.1575463	9
8	Absolut	15	2002	4475	0	594000	132.7374	30387.3	424.8	523.1	1	30812.1	1	59589	0	0.1504252	10
9	Absolut	15	2003	4488	0	620000	138.1462	29275	958.6	2819.4	315.5	30233.6	1	59608	0	0.1392448	11
10	Absolut	15	2004	4640	0	658880	142	31208.5	5.9	2185.1	2202.4	31214.4	1	59488	0	0.1343526	12

The dataset offers a comprehensive brand performance analysis, encompassing sales, and advertising expenditure.

Variables and Their Descriptions:

- **BrandName:** This variable is of character type with a length of 9. It stores the name of the brand.
- **Brand_ID:** Numeric type variable with a length of 8, used to uniquely identify each brand.
- **Broad:** Numeric type variable with a length of 8, which refers to broadcast advertising expenditure.
- **DollarSales:** This is a numeric type of variable with a length of 8, likely indicating sales in dollar terms. It is the product $\text{Priceperunit} * \text{TotalSales}$.
- **Domestic:** A numeric type variable with a length of 8, serves as a dummy variable indicating if the brand is domestic (1) or international (0).
- **FirstIntro:** Numeric type variable with a length of 8, acts as a dummy variable capturing whether the brand is in the first three years of its introduction (1 for yes, 0 otherwise).
- **Mag:** Numeric type variable with a length of 8, referring to magazine advertising expenditure.
- **Marketshare:** Numeric type variable with a length of 8, indicating the market share of the brand.
- **News:** Numeric type variable with a length of 8, referring to newspaper advertising expenditure.
- **Outdoor:** Numeric type variable with a length of 8, referring to outdoor advertising expenditure.
- **PricePerUnit:** Numeric type variable with a length of 8, denoting the price of a 9-liter case.
- **Print:** Numeric type variable with a length of 8, representing the sum of "Mag" and "News" advertising expenditures.
- **Tier:** Numeric type variable with a length of 8, used to categorize the brand into Tier 1, 2, or 3.
- **TotalMinusSales:** Numeric type variable with a length of 8, indicating total sales in the industry minus the brand's own sales.
- **TotalSales:** Numeric type variable with a length of 8, indicating the total quantity of 9-liter cases sold for the brand.
- **Year:** Numeric type variable with a length of 8, representing the year of observation.
- **YearID:** Numeric type variable with a length of 8, serving as a unique identifier for each year.

2. Summary Statistics of the Data

The SAS System

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
TotalSales	263	1400.41	1597.96	42.0000000	9015.00
Domestic	263	0.6121673	0.4881851	0	1.0000000
DollarSales	263	113912.88	166365.75	4690.00	786721.00
PricePerUnit	263	83.7111141	57.1260813	27.2912300	250.2623000
Mag	263	3280.29	6690.51	1.0000000	33971.30
News	263	193.9098859	549.9240691	0.3000000	3524.90
Outdoor	263	162.0410646	467.1365724	1.0000000	3255.40
Broad	263	275.9220532	1043.66	0.6000000	7827.10
Print	263	3474.20	6878.26	2.0000000	34504.70
Tier	263	2.2889734	0.8196956	1.0000000	3.0000000
TotalMinusSales	263	62653.85	1599.77	55209.00	64163.00
Firstintro	263	0.0152091	0.1226172	0	1.0000000
Marketshare	263	0.0469717	0.0538309	0.0014675	0.2704770

There are in total **263 observations** in the dataset and there are **no missing values** in the dataset.

- **Total Sales** - The average number of 9L cases sold across brands is 1400, but we also have a standard deviation of 1597.96. There seems to be high variability among brands. Some brands can sell very few cases whereas others can sell more cases.
- **Domestic** – On average, 62% of the brands are domestic. With the standard deviation given which is 0.488, there appears to be high variability. Er can infer that the market is majorly dominated by the local brand, although, the international brand still has a strong presence in the market by looking.
- **Dollar Sales** – The average revenue generated by the brand is \$113912.88, with a standard deviation of \$166365.75 which is 146.04% of the average revenue generated by brands, reflecting a very high variability among brands. It indicates that some brands have generated very high revenue and some generated very low revenue. It seems some brands have outperformed or underperformed.
- **Price per unit** – On average, the price of the 9L case is \$83.711 $\approx$ \$84, with a standard deviation of \$57.12 which is showing 68% of the variability. The minimum

price is \$27.29, and the maximum price is \$250.26. This column suggests that we have a mix of brands that are premium as well as budget - friendly.

- **Mag** – On average, \$3280.29 was spent on the magazine advertisement on brands with a standard deviation of \$6690.51 which is 203.9% of the variability explained. With so much high variability, we can infer that some brands are spending a lot on magazines, and some are spending less with the highest they spent is \$33971.30.
- **News** – On average, \$162.04 was spent on news advertising, with a standard deviation of \$549.92 which is 339.37% of the variability which is also very high variability. We can infer that some brands are investing a lot in news advertising, and some are investing less. The maximum they spent on news advertising was \$3524.00.
- **Outdoor** – On average, they invested \$162.04 on outdoor advertising, with a standard deviation of \$467.13, which is 288.28% of the variability, which is a very high variability explained. We can infer that some brands have invested a lot in outdoor advertising and some brands invested very less in outdoor advertising, with a maximum they invested \$3255.40.
- **Broad** – On average, \$275 was invested in broadcast advertising, with a standard deviation of \$1043.66, which is 379.51% of the variability explained by the data, which is very high. We can infer that some brands have invested a lot in broadcast advertising and some brands invested very less in outdoor advertising, with a maximum they invested \$7827.

The SAS System

The FREQ Procedure

BrandName	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Absolut	13	4.94	13	4.94
Aristocra	13	4.94	26	9.89
Barton	13	4.94	39	14.83
Belvedere	7	2.66	46	17.49
Burnett	11	4.18	57	21.67
Chopin	7	2.66	64	24.33
Crystal P	13	4.94	77	29.28
Finlandia	13	4.94	90	34.22
Fleischma	12	4.56	102	38.78
Fris	10	3.80	112	42.59
Gilbey's	13	4.94	125	47.53
Gordon's	13	4.94	138	52.47
Grey Goos	9	3.42	147	55.89
Kamchatka	13	4.94	160	60.84
Ketel One	8	3.04	168	63.88
Level	2	0.76	170	64.64
McCormick	10	3.80	180	68.44
Polar Ice	3	1.14	183	69.58
Popov	13	4.94	196	74.52
Pravda	2	0.76	198	75.29
Seagram's	1	0.38	199	75.67
Skol	13	4.94	212	80.61
Sky	10	3.80	222	84.41
Smirnoff	13	4.94	235	89.35
Stolichnay	13	4.94	248	94.30
Tanqueray	9	3.42	257	97.72
Three Oll	6	2.28	263	100.00

Tier	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	61	23.19	61	23.19
2	65	24.71	126	47.91
3	137	52.09	263	100.00

- **Tier** – From the frequency table provided, most of the brands belong to Tier - 3 by up to 52.09% followed by Tier -2 by up to 24.71%.

Pearson Correlation Coefficients, N = 263 Prob > r under H0: Rho=0										
	TotalSales	PricePerUnit	Print	Domestic	Outdoor	Broad	Firstintro	Marketshare	TotalMinusSales	Tier
TotalSales	1.00000	-0.03648 0.5559	0.47663 <.0001	0.15334 0.0128	0.54368 <.0001	0.66012 <.0001	-0.10078 0.1029	0.96411 <.0001	-0.99735 <.0001	-0.16751 0.0065
PricePerUnit	-0.03648 0.5559	1.00000	0.43494 <.0001	-0.83153 <.0001	0.19332 0.0016	0.19765 0.0013	0.13658 0.0268	-0.07845 0.2047	0.05315 0.3906	-0.89588 <.0001
Print	0.47663 <.0001	0.43494 <.0001	1.00000	-0.43594 <.0001	0.60462 <.0001	0.18551 0.0025	0.00004 0.9994	0.49432 <.0001	-0.47317 <.0001	-0.61947 <.0001
Domestic	0.15334 0.0128	-0.83153 <.0001	-0.43594 <.0001	1.00000	-0.19846 0.0012	-0.06157 0.3199	-0.15613 0.0112	0.17158 0.0053	-0.16256 0.0083	0.86296 <.0001
Outdoor	0.54368 <.0001	0.19332 0.0016	0.60462 <.0001	-0.19846 0.0012	1.00000	0.30346 <.0001	0.17393 0.0047	0.53097 <.0001	-0.53859 <.0001	-0.37307 <.0001
Broad	0.66012 <.0001	0.19765 0.0013	0.18551 0.0025	-0.06157 0.3199	0.30346 <.0001	1.00000	-0.03281 0.5963	0.49493 <.0001	-0.64171 <.0001	-0.25089 <.0001
Firstintro	-0.10078 0.1029	0.13658 0.0268	0.00004 0.9994	-0.15613 0.0112	0.17393 0.0047	-0.03281 0.5963	1.00000	-0.10108 0.1019	0.10454 0.0907	-0.15782 0.0104
Marketshare	0.96411 <.0001	-0.07845 0.2047	0.49432 <.0001	0.17158 0.0053	0.53097 <.0001	0.49493 <.0001	-0.10108 0.1019	1.00000	-0.97438 <.0001	-0.14363 0.0198
TotalMinusSales	-0.99735 <.0001	0.05315 0.3906	-0.47317 <.0001	-0.16256 0.0083	-0.53859 <.0001	-0.64171 <.0001	0.10454 0.0907	-0.97438 <.0001	1.00000	0.15578 0.0114
Tier	-0.16751 0.0065	-0.89588 <.0001	-0.61947 <.0001	0.86296 <.0001	-0.37307 <.0001	-0.25089 <.0001	-0.15782 0.0104	-0.14363 0.0198	0.15578 0.0114	1.00000

Correlation Analysis

- ➔ We can infer from the correlation matrix that there seems to be a positive correlation between print advertising, focusing a lot on domestic brands, outdoor advertising, and broadcast advertising concerning total sales. If we increase our investment in print advertising, then there will be an increase in total sales of up to 47.6%. If we increase our investment in outdoor advertising, then there will be an increase in total sales of up to 54%. Similarly, if we increase our investment in broadcast advertising, we can increase by sales up to 66%. If we target a lot of local brands, sales will also increase by 15%. These variables are highly significant as their p-value is less than 0.05.
- ➔ If we increase our price per unit, there will be a decrease in total sales of 3%. The variable is not significant as its p-value is more than 0.05. We can infer that price per unit alone is not an important variable if we want to consider how it impacts total sales. It will not have a statistical significance alone in the total sales.

3. Linear Model I

Run a linear regression of sales on prices. Write out the regression equation. What can you conclude about Vodka prices?

The SAS System					
The REG Procedure					
Model: MODEL1					
Dependent Variable: TotalSales					
Number of Observations Read		263			
Number of Observations Used		263			
Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	890287	890287	0.35	0.5559
Error	261	668119472	2559845		
Corrected Total	262	669009760			
Root MSE		1599.95147	R-Square	0.0013	
Dependent Mean		1400.41445	Adj R-Sq	-0.0025	
Coeff Var		114.24843			
Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1485.83515	175.25254	8.48	<.0001
PricePerUnit	1	-1.02042	1.73030	-0.59	0.5559

➔ The Regression equation is as follows:

$$\text{TotalSales} = 1485.83515 - 1.02042 \times \text{PricePerUnit} + \varepsilon$$

Explanation: 1 unit increase in price per unit reduces total sales by -1.02042 units.

- ➔ Analysis of Variance p-value is 0.5559 more than 0.05, hence the model is not significant.
- ➔ The price Per Unit p-value is also 0.5559 which is more than 0.05, hence the variable itself is not significant enough to explain if it has any impact on total sales.
- ➔ $R^2 = 0.0013$ which is very low. It explains that only 1.3% of the variability is explained by the model.

4. Linear Model II

Run a regression that includes price and advertising. What variables in the data represent advertising? Describe your methodology for choosing your model specification. Write out the regression equations for your specification. Comment on your results.

The SAS System

The REG Procedure

Model: MODEL1

Dependent Variable: TotalSales

Number of Observations Read	263
Number of Observations Used	263

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	473557861	94711572	124.54	<.0001
Error	257	195451899	760513		
Corrected Total	262	669009760			

Root MSE	872.07409	R-Square	0.7078
Dependent Mean	1400.41445	Adj R-Sq	0.7022
Coeff Var	62.27257		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1582.55091	96.85015	16.34	<.0001
PricePerUnit	1	-10.47726	1.09821	-9.54	<.0001
Broad	1	0.92121	0.05492	16.77	<.0001
Outdoor	1	0.61292	0.15163	4.04	<.0001
Mag	1	0.09654	0.01092	8.84	<.0001
News	1	0.12754	0.10812	1.18	0.2392

- Variables in the data that represent advertising are Broad, Outdoor, Mag, and News.
- The methodology for choosing model specifications was to choose Mag and News instead of Print. Print is the summation of Mag and news and if we would use just print, we wouldn't have any detailed information on which print media is having more impact on total sales variable.
- Regression equation is as follows:

$$\text{TotalSales} = 1582.55091 - 10.47726 \times \text{PricePerUnit} + 0.92121 \times \text{Broad} + 0.61292 \times \text{Outdoor} + 0.09654 \times \text{Mag} + 0.12754 \times \text{News} + \varepsilon$$

- One unit increase in price per unit decreases total sales by -10.47726.
 - One unit increase in broadcast advertising investment increases total sales by 0.92121.
 - One unit increase in outdoor advertising investment increases total sales by 0.61292.
 - One unit increase in magazine advertising investment increases total sales by 0.09654.
 - One unit increase in news advertising investment increases total sales by 0.12754.
-
- $R^2 = 0.7078$ & Adjusted $- R^2 = 0.7022$. We can infer that **70.22%** of the variance in the total sales is explained by the independent variables.
 - Analysis of variance p-value is less than 0.05, we can infer that the model is highly significant.
 - Among the independent variables, except News, all other variables are highly significant because other variables' p-values are less than 0.05 wherein the p-value of news is more than 0.05 which is 0.2392. We can infer that investing in news advertising does not have a statistically significant impact on our total sales.

5. Encoding Dummy Variables

The dataset classifies the brands into three quality tiers.

- ➔ Tier 1 brands: Chopin, Belvedere, Grey Goose, Level, Ketel One, Absolut, Stolichnaya, Pravda, and Ciroc.
- ➔ Tier 2 brands: Smirnoff, Sky, Finlandia, Three Olives, Fris, Seagrams, Polar Ice, Van Gogh, Tanqueray, Pearl, Iceburg, and Svedka
- ➔ Tier 3 brands: All remaining brands

Create dummy variables for the Tier variable. How many dummy variables will you need?

There are 3 tiers in our data, hence 3 categories. Then, the number of dummy variables is $3 - 1 = 2$. **We will be creating two dummy variables.** They're dummy_tier1 and dummy_tier2.

The SAS System

Obs	BrandName	Brand_ID	Year	TotalSales	Domestic	DollarSales	PricePerUnit	Mag	News	Outdoor	Broad	Print	Tier	TotalMinusSales	Firstintro	Marketshare	YearID	dummy_tier1	dummy_tier2
1	Absolut	15	1995	3000	0	352040	117.3467	25713.7	291.4	332.7	1	26005.1	1	60840	0	0.1339226	3	1	0
2	Absolut	15	1996	3340	0	370072	110.8	25658	177.6	625	1	25835.6	1	60532	0	0.1420068	4	1	0
3	Absolut	15	1997	3440	0	381152	110.8	27013.9	201.7	515.7	1	27215.6	1	60464	0	0.1467577	5	1	0
4	Absolut	15	1998	3630	0	419737	115.63	27617	308.2	1420.4	161.9	27925.2	1	60306	0	0.1461059	6	1	0
5	Absolut	15	1999	4050	0	468302	115.6301	30605.3	258.1	1416.8	70.3	30863.4	1	59918	0	0.1612839	7	1	0
6	Absolut	15	2000	4605	0	560025	121.6124	33971.3	533.4	1336.7	143.1	34504.7	1	59395	0	0.173368	8	1	0
7	Absolut	15	2001	4379	0	556550	127.0952	28559.8	231.3	1999	134.2	28791.1	1	59653	0	0.1575463	9	1	0
8	Absolut	15	2002	4475	0	594000	132.7374	30387.3	424.8	523.1	1	30812.1	1	59589	0	0.1504252	10	1	0
9	Absolut	15	2003	4488	0	620000	138.1462	29275	958.6	2819.4	315.5	30233.6	1	59608	0	0.1392448	11	1	0
10	Absolut	15	2004	4640	0	658880	142	31208.5	5.9	2185.1	2202.4	31214.4	1	59488	0	0.1343526	12	1	0

6. Linear Model III

Run a regression model to understand the influence of vodka quality tiers on sales by including Tier dummies in the above model. What do the results tell you about Tier 1 and Tier 2 brands? Should Svedka management consider competing in Tier 1?

The SAS System

The REG Procedure
Model: MODEL1
Dependent Variable: TotalSales

Number of Observations Read	263
Number of Observations Used	263

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	478642307	68377472	91.59	<.0001
Error	255	190367453	746539		
Corrected Total	262	669009760			

Root MSE	864.02490	R-Square	0.7154
Dependent Mean	1400.41445	Adj R-Sq	0.7076
Coeff Var	61.69780		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1660.15883	117.25416	14.16	<.0001
PricePerUnit	1	-14.23154	2.29325	-6.21	<.0001
Broad	1	0.90289	0.05487	16.45	<.0001
Outdoor	1	0.54541	0.15245	3.58	0.0004
Mag	1	0.09025	0.01202	7.51	<.0001
News	1	0.09987	0.10992	0.91	0.3644
dummy_tier1	1	630.47593	374.99529	1.68	0.0939
dummy_tier2	1	535.89530	210.37597	2.55	0.0114

➔ Regression equation is as follows:

$$\begin{aligned} \text{TotalSales} = & 1660.15883 - 14.23154 \times \text{PricePerUnit} + 0.90289 \times \text{Broad} \\ & + 0.54541 \times \text{Outdoor} + 0.09025 \times \text{Mag} + 0.09987 \times \text{News} \\ & + 630.47593 \times \text{dummytier1} + 535.89530 \times \text{dummytier2} + \epsilon \end{aligned}$$

- One unit increase in price per unit decreases total sales by -14.23154.
 - One unit increase in investment in broadcast advertisement increases total sales by 0.90289.
 - One unit increase in investment in outdoor advertising increases total sales by 0.54541.
 - One unit increase in investment in magazine advertising increases total sales by 0.09025.
 - One unit increase in investment in news advertising increases total sales by 0.09987.
 - If the brand belongs to tier 1, then total sales will increase by 630.47593 units.
 - If the brand belongs to tier 2, then total sales will increase by 535.89530 units.
-
- ➔ Variable “News” is not significant as its p-value is 0.3644 which is much higher than 0.05. Hence, investing in news advertising will not have a statistically meaningful impact on total sales.
 - ➔ Similarly, dummy_tier1 is not statistically significant as its p-value is 0.0939 which is comparatively higher than 0.05.
 - ➔ Analysis of Variance p-value is less than 0.05, hence, we can say that this model is statistically significant and hence useful in predicting total sales.
 - ➔ $R^2 = 0.7154$ & Adjusted $-R^2 = 0.7076$. We can infer that 70.76% of the variance in the total sales is explained by the independent variables.

Should Svedka management consider investing in Tier 1?

- ➔ Svedka management should not consider investing in tier 1. Although investing in tier 1 has more potential for sales growth as it is associated with a 630-unit increase in sales growth. However, it's not statistically significant as its p-value is less than 0.05. Hence, we don't have enough evidence to guarantee sales growth as there's risk involved in investing there.
- ➔ Svedka should invest in tier 2 which has significant potential for growth in sales as it's statistically significant.

7. Computing Price Elasticity of Demand

Compute the demand elasticity of price using the estimates from the chosen model from the previous questions. What can you conclude about Vodka prices based on the elasticity measures?

Regression Models			
	Model 1	Model 2	Model 3
DV	Total Sales	Total Sales	Total Sales
Intercept	1485.83515 (<0.0001)	1582.55091 (<0.0001)	1660.15883 (<0.0001)
Priceperunit	-1.02042 (-0.5559)	-10.47726 (<0.0001)	-14.23154 (<0.0001)
Broad		0.92121 (<0.0001)	0.90289 (<0.0001)
Outdoor		0.61292 (<0.0001)	0.54541 (0.0004)
Mag		0.09694 (<0.0001)	0.09025 (<0.0001)
New		0.12754 (-0.2392)	0.09987 (0.3644)
dummy_tier1			630.47593 (0.0939)
dummy_tier2			535.8953 0.0114
R-square	0.0013	0.7078	0.7154
Adj- R square	-0.0025	0.7022	0.7076

➔ Based on the models we have; our ideal model is **Model 3** as it has a higher Adjusted – R square as compared to other models. Since it is the linear model, the formula for the price elasticity of demand is as follows:

➔ Price coefficient (β_i) from our chosen model = -14.23154

$$\text{Price Elasticity Of Demand} = \beta_i \times \frac{\text{Mean(Price}_i\text{)}}{\text{Mean(Sales}_i\text{)}} = -14.23154 \times \frac{83.7111141}{1400.41} = -0.85$$

➔ The price elasticity of demand is -0.85. The negative sign indicates that if the price increases, demand decreases. One percent increase in price per unit decreases the total sales by 0.85%. Demand is inelastic as the price elasticity of demand is less than 1%. It means that the change in price per unit has little effect on total sales. Total sales are not price-sensitive.

8. Recommendations for Svedka

Based on your analysis of the Vodka data, what recommendations can you make to the SVEDKA management. Specifically what recommendations would you make based on estimated Price, Advertisement and Tier 1 coefficients. Based on your analysis of the demand elasticity, what would you recommend to the management?

Recommendations for Svedka

- ➔ The price elasticity of demand is -0.85 which is inelastic. Our sales seem to be not highly price sensitive. Hence, consider increasing the pricing of the product to a moderate level to boost sales.
- ➔ Consider investing in broadcast advertising, outdoor advertising and magazine advertising as they have a positive impact on sales growth as they're statistically significant enough to prove that they can lead to sales growth. Reduce the allocation of the advertising budget to News advertising. Invest more in broadcast advertising as it has a more positive impact on sales (0.90289).
- ➔ Focus on competing in tier 2, as it is evident that tier 2 brands have a statistically significant impact on sales of the brand. Evaluate carefully for tier 1 before competing as it possesses a certain risk.

APPENDIX

```

* 1. Convert the data file SVEDKA.csv to a SAS dataset. Study the data along with the data dictionary and write a summary about it.;
libname assignmt1 '\\apporto.com\dfs\IIT\Users\gauravsinha_iit\Desktop\Quantitative Marketing Models';

❏proc import out= assignmt1.svedkadata
    datafile= "\\apporto.com\dfs\IIT\Users\gauravsinha_iit\Desktop\Quantitative Marketing Models\Assignment 1\svedka.csv"
    dbms= csv;
    getnames=yes;
    datarow=2;
run;

/* looking at the content of the dataset imported */;

❏proc contents data= assignmt1.svedkadata;
run;

/* printing some of the records of the dataset */;

❏proc print data=assignmt1.svedkadata (obs=10);
run;

* 2. Generate summary statistics for the data. What can you infer about the data at this stage?;

❏proc means data = assignmt1.svedkadata;
var TotalSales Domestic DollarSales PricePerUnit Mag News Outdoor Broad Print Tier TotalMinusSales Firstintro Marketshare;
run;

❏proc freq data = assignmt1.svedkadata;
table brandname tier;
run;

❏proc corr data = assignmt1.svedkadata;
var totalsales priceperunit print domestic outdoor broad firstintro marketshare totalminussales tier;
run;

*3. Run a linear regression of sales on prices. Write out the regression equation. What can you conclude about Vodka prices?;
/* Model 1 */
❏proc reg data = assignmt1.svedkadata plots=none;
model totalsales = priceperunit;
run; quit;

* 4. Run a regression which includes price and advertising. What variables in the data represent advertising?
Describe your methodology for choosing your model specification. Write out the regression equations for your specification.
Comment on your results. ;
/* Model 2 */
❏proc reg data = assignmt1.svedkadata plots = none;
model totalsales = priceperunit broad outdoor mag news;
run; quit;

* 5. The dataset classifies the brands into three quality tiers.
Tier 1 brands: Chopin, Belvedere, Grey Goose, Level, Ketel One, Absolut, Stolichnaya, Pravda, and Ciroc.
Tier 2 brands: Smirnoff, Sky, Finlandia, Three Olives, Fris, Seagrams, Polar Ice, Van Gogh, Tanqueray, Pearl, Iceburg, and Svedka
Tier 3 brands: All remaining brands

Create dummy variables for the Tier variable. How many dummy variables will you need?;

❏data assignmt1.svedkadata_updated;
set assignmt1.svedkadata;
dummy_tier1 = (Tier = 1);
dummy_tier2 = (Tier = 2);
run;

❏proc print data = assignmt1.svedkadata_updated (obs=10);
run;

* 6. Run a regression model to understand the influence of vodka quality tiers on sales by including Tier dummies to the above model.
What do the results tell you about Tier 1 and Tier 2 brands? Should Svedka management consider competing in Tier 1?;

/* Model 3 */
❏proc reg data = assignmt1.svedkadata_updated plots = none;
model totalsales = priceperunit broad outdoor mag news dummy_tier1 dummy_tier2;
run; quit;

```